Resampling Techniques, Bootstrap and Blocking

Morten Hjorth-Jensen Email morten.hjorth-jensen@fys.uio.no[1]

Department of Physics and Center fo Computing in Science Education, University of Oslo, Oslo, Norway[1]

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Overview of week March 16-20, 2025

Topics

1. Resampling Techniques, Bootstrap and Blocking
2. Calculation of one-body densities (see whiteboard notes)
3. Video of lecture
4. Handwritten notes

Why resampling methods ?

Statistical analysis

- Our simulations can be treated as *computer experiments*. This is particularly the case for Monte Carlo methods
- The results can be analysed with the same statistical tools as we would use analysing experimental data.
- As in all experiments, we are looking for expectation values and an estimate of how accurate they are, i.e., possible sources for errors.

Statistical analysis

- As in other experiments, many numerical experiments have two classes of errors:
  1. Statistical errors
  2. Systematical errors
- Statistical errors can be estimated using standard tools from statistics
- Systematical errors are method specific and must be treated differently from case to case.

And why do we use such methods?

As you will see below, due to correlations between various measurements, we need to evaluate the so-called covariance in order to establish a proper evaluation of the total variance and the thereby the standard deviation of a given expectation value.
The covariance however, leads to an evaluation of a double sum over the various stochastic variables. This becomes computationally too expensive to evaluate. Methods like the Bootstrap, the Jackknife and/or Blocking allow us to circumvent this problem.

Central limit theorem, reminder from last week

Last week we derived the central limit theorem with the following assumptions:

Measurement i

We assumed that each individual measurement x_{ij} is represented by stochastic variables which independent and identically distributed (iid). This defined the sample mean of of experiment i with n samples as

$$\overline{x}_i = \frac{1}{n} \sum_j x_{ij}.$$

and the sample variance

$$\sigma_i^2 = \frac{1}{n} \sum_j \left(x_{ij} - \overline{x}_i\right)^2.$$

Further remarks

Note that we use n instead of $n - 1$ in the definition of variance. The sample variance and the sample mean are not necessarily equal to the exact values we would get if we knew the corresponding probability distribution.

Statistics, more technicalities

The desired variance $\mathrm{var}(\overline{X}_n)$, i.e. the sample error squared err_X^2, is given by:

$$\mathrm{err}_X^2 = \mathrm{var}(\overline{X}_n) = \frac{1}{n^2} \sum_{ij} \mathrm{cov}(X_i, X_j) \tag{1}$$

We see now that in order to calculate the exact error of the sample with the above expression, we would need the true means μ_{X_i} of the stochastic variables X_i. To calculate these requires that we know the true multivariate PDF of all the X_i. But this PDF is unknown to us, we have only got the measurements of one sample. The best we can do is to let the sample itself be an estimate of the PDF of each of the X_i, estimating all properties of X_i through the measurements of the sample.

Statistics

Our estimate of μ_{X_i} is then the sample mean $\bar{x}$ itself, in accordance with the the central limit theorem:

$$\mu_{X_i} = \langle x_i \rangle \approx \frac{1}{n} \sum_{k=1}^{n} x_k = \bar{x}$$

Using $\bar{x}$ in place of μ_{X_i} we can give an *estimate* of the covariance in Eq. (1)

$$\mathrm{cov}(X_i, X_j) = \langle (x_i - \langle x_i \rangle)(x_j - \langle x_j \rangle) \rangle \approx \langle (x_i - \bar{x})(x_j - \bar{x}) \rangle,$$

resulting in

$$\frac{1}{n} \sum_{l}^{n} \left(\frac{1}{n} \sum_{k}^{n} (x_k - \bar{x}_n)(x_l - \bar{x}_n) \right) = \frac{1}{n} \frac{1}{n} \sum_{kl} (x_k - \bar{x}_n)(x_l - \bar{x}_n) = \frac{1}{n} \mathrm{cov}(x)$$

Statistics and sample variance

By the same procedure we can use the sample variance as an estimate of the variance of any of the stochastic variables X_i

$$\mathrm{var}(X_i) = \langle x_i - \langle x_i \rangle \rangle \approx \langle x_i - \bar{x}_n \rangle,$$

which is approximated as

$$\mathrm{var}(X_i) \approx \frac{1}{n} \sum_{k=1}^{n} (x_k - \bar{x}_n) = \mathrm{var}(x) \tag{2}$$

Now we can calculate an estimate of the error err_X of the sample mean $\bar{x}_n$:

$$\begin{aligned} \mathrm{err}_X^2 &= \frac{1}{n^2} \sum_{ij} \mathrm{cov}(X_i, X_j) \\ &\approx \frac{1}{n^2} \sum_{ij} \frac{1}{n} \mathrm{cov}(x) = \frac{1}{n^2} n^2 \frac{1}{n} \mathrm{cov}(x) \\ &= \frac{1}{} \end{aligned}$$

Statistics, uncorrelated results

In the special case that the measurements of the sample are uncorrelated (equivalently the stochastic variables X_i are uncorrelated) we have that the off-diagonal elements of the covariance are zero. This gives the following estimate of the sample error:

$$\mathrm{err}_X^2 = \frac{1}{n^2}\sum_{ij}\mathrm{cov}(X_i, X_j) = \frac{1}{n^2}\sum_{i}\mathrm{var}(X_i),$$

resulting in

$$\mathrm{err}_X^2 \approx \frac{1}{n^2}\sum_{i}\mathrm{var}(x) = \frac{1}{n}\mathrm{var}(x) \tag{4}$$

where in the second step we have used Eq. (2). The error of the sample is then just its standard deviation divided by the square root of the number of measurements the sample contains. This is a very useful formula which is easy to compute. It acts as a first approximation to the error, but in numerical experiments, we cannot overlook the always present correlations.

Statistics, computations

For computational purposes one usually splits up the estimate of err_X^2, given by Eq. (3), into two parts

$$\mathrm{err}_X^2 = \frac{1}{n}\mathrm{var}(x) + \frac{1}{n}(\mathrm{cov}(x) - \mathrm{var}(x)),$$

which equals

$$\frac{1}{n^2}\sum_{k=1}^{n}(x_k - \bar{x}_n)^2 + \frac{2}{n^2}\sum_{k<l}(x_k - \bar{x}_n)(x_l - \bar{x}_n) \tag{5}$$

The first term is the same as the error in the uncorrelated case, Eq. (4). This means that the second term accounts for the error correction due to correlation between the measurements. For uncorrelated measurements this second term is zero.

Statistics, more on computations of errors

Computationally the uncorrelated first term is much easier to treat efficiently than the second.

$$\mathrm{var}(x) = \frac{1}{n}\sum_{k=1}^{n}(x_k - \bar{x}_n)^2 = \left(\frac{1}{n}\sum_{k=1}^{n} x_k^2\right) - \bar{x}_n^2$$

We just accumulate separately the values x^2 and x for every measurement x we receive. The correlation term, though, has to be calculated at the end of the experiment since we need all the measurements to calculate the cross terms. Therefore, all measurements have to be stored throughout the experiment.

Statistics, wrapping up 1

Let us analyze the problem by splitting up the correlation term into partial sums of the form:

$$f_d = \frac{1}{n-d} \sum_{k=1}^{n-d} (x_k - \bar{x}_n)(x_{k+d} - \bar{x}_n)$$

The correlation term of the error can now be rewritten in terms of f_d

$$\frac{2}{n} \sum_{k<l} (x_k - \bar{x}_n)(x_l - \bar{x}_n) = 2 \sum_{d=1}^{n-1} f_d$$

The value of f_d reflects the correlation between measurements separated by the distance d in the sample samples. Notice that for $d = 0$, f is just the sample variance, $\mathrm{var}(x)$. If we divide f_d by $\mathrm{var}(x)$, we arrive at the so called *autocorrelation function*

$$\kappa_d = \frac{f_d}{\mathrm{var}(x)}$$

Statistics, final expression

The sample error (see eq. (5)) can now be written in terms of the autocorrelation function:

$$\begin{aligned}
\mathrm{err}_X^2 &= \frac{1}{n}\mathrm{var}(x) + \frac{2}{n}\cdot\mathrm{var}(x)\sum_{d=1}^{n-1}\frac{f_d}{\mathrm{var}(x)} \\
&= \left(1 + 2\sum_{d=1}^{n-1}\kappa_d\right)\frac{1}{n}\mathrm{var}(x) \\
&= \frac{\tau}{n}\cdot\mathrm{var}(x) \qquad (6)
\end{aligned}$$

and we see that err_X can be expressed in terms the uncorrelated sample variance times a correction factor τ which accounts for the correlation between measurements. We call this correction factor the *autocorrelation time*:

$$\tau = 1 + 2\sum_{d=1}^{n-1}\kappa_d \qquad (7)$$

Statistics, effective number of correlations

For a correlation free experiment, τ equals 1. From the point of view of eq. (6) we can interpret a sequential correlation as an effective reduction of the number of measurements by a factor τ. The effective number of measurements becomes:

$$n_{\mathrm{eff}} = \frac{n}{\tau}$$

To neglect the autocorrelation time τ will always cause our simple uncorrelated estimate of $\mathrm{err}_X^2 \approx \mathrm{var}(x)/n$ to be less than the true sample error. The estimate of the error will be too *good*. On the other hand, the calculation of the full autocorrelation time poses an efficiency problem if the set of measurements is very large.

Computing the correlation function

This code is best seen with the jupyter-notebook

```
#!/usr/bin/env python
import numpy as np
import matplotlib.mlab as mlab
import matplotlib.pyplot as plt
import random

# initialize the rng with a seed, simple uniform distribution
random.seed()
m = 10000
samplefactor = 1.0/m
x = np.zeros(m)
MeanValue = 0.
VarValue = 0.
for i in range (m):
    value = random.random()
    x[i] = value
    MeanValue += value
    VarValue += value*value

MeanValue *= samplefactor
VarValue *= samplefactor
Variance = VarValue-MeanValue*MeanValue
STDev = np.sqrt(Variance)
print("MeanValue =", MeanValue)
print("Variance =", Variance)
print("Standard deviation =", STDev)

# Computing the autocorrelation function
```

Time Auto-correlation Function

The so-called time-displacement autocorrelation $\phi(t)$ for a quantity M is given by

$$\phi(t) = \int dt' \left[\mathrm{M}(t') - \langle \mathrm{M} \rangle\right] \left[\mathrm{M}(t'+t) - \langle \mathrm{M} \rangle\right],$$

which can be rewritten as

$$\phi(t) = \int dt' \left[\mathrm{M}(t')\mathrm{M}(t'+t) - \langle \mathrm{M} \rangle^2\right],$$

where $\langle \mathrm{M} \rangle$ is the average value and $\mathrm{M}(t)$ its instantaneous value. We can discretize this function as follows, where we used our set of computed values $\mathrm{M}(t)$ for a set of discretized times (our Monte Carlo cycles corresponding to moving all electrons?)

$$\phi(t) = \frac{1}{t_{\max} - t} \sum_{t'=0}^{t_{\max}-t} \mathrm{M}(t')\mathrm{M}(t'+t) - \frac{1}{t_{\max} - t} \sum_{t'=0}^{t_{\max}-t} \mathrm{M}(t') \times \frac{1}{t_{\max} - t} \sum_{t}^{t_{\mathrm{m}}}$$

Time Auto-correlation Function

One should be careful with times close to $t_{\max}$, the upper limit of the sums becomes small and we end up integrating over a rather small time interval. This means that the statistical error in $\phi(t)$ due to the random nature of the fluctuations in M(t) can become large. One should therefore choose $t \ll t_{\max}$.
Note that the variable M can be any expectation values of interest. The time-correlation function gives a measure of the correlation between the various values of the variable at a time t' and a time $t' + t$. If we multiply the values of M at these two different times, we will get a positive contribution if they are fluctuating in the same direction, or a negative value if they fluctuate in the opposite direction. If we then integrate over time, or use the discretized version of, the time correlation function $\phi(t)$ should take a non-zero value if the fluctuations are correlated, else it should gradually go to zero. For times a long way apart the different values of M are most likely uncorrelated and $\phi(t)$ should be zero.

Time Auto-correlation Function

We can derive the correlation time by observing that our Metropolis algorithm is based on a random walk in the space of all possible spin configurations. Our probability distribution function $\hat{\mathrm{w}}(t)$ after a given number of time steps t could be written as

$$\hat{\mathrm{w}}(t) = \hat{\mathrm{W}}^{\mathrm{t}}\hat{\mathrm{w}}(0),$$

with $\hat{\mathrm{w}}(0)$ the distribution at $t = 0$ and $\hat{\mathrm{W}}$ representing the transition probability matrix. We can always expand $\hat{\mathrm{w}}(0)$ in terms of the right eigenvectors of $\hat{\mathrm{v}}$ of $\hat{\mathrm{W}}$ as

$$\hat{\mathrm{w}}(0) = \sum_i \alpha_i \hat{\mathrm{v}}_i,$$

resulting in

$$\hat{\mathrm{w}}(t) = \hat{\mathrm{W}}^t\hat{\mathrm{w}}(0) = \hat{\mathrm{W}}^t \sum_i \alpha_i \hat{\mathrm{v}}_i = \sum_i \lambda_i^t \alpha_i \hat{\mathrm{v}}_i,$$

Time Auto-correlation Function

If we assume that λ_0 is the largest eigenvector we see that in the limit $t \to \infty$, $\hat{\mathrm{w}}(t)$ becomes proportional to the corresponding eigenvector $\hat{\mathrm{v}}_0$. This is our steady state or final distribution.
We can relate this property to an observable like the mean energy. With the probabilty $\hat{\mathrm{w}}(t)$ (which in our case is the squared trial wave function) we can write the expectation values as

$$\langle \mathrm{M}(t) \rangle = \sum_{\mu} \hat{\mathrm{w}}(t)_{\mu} \mathrm{M}_{\mu},$$

or as the scalar of a vector product

$$\langle \mathrm{M}(t) \rangle = \hat{\mathrm{w}}(t) \mathrm{m},$$

with m being the vector whose elements are the values of M_{μ} in its various microstates μ.

Time Auto-correlation Function

We rewrite this relation as

$$\langle \mathrm{M}(t) \rangle = \hat{\mathrm{w}}(t)\mathrm{m} = \sum_i \lambda_i^t \alpha_i \hat{\mathrm{v}}_i \mathrm{m}_i.$$

If we define $m_i = \hat{\mathrm{v}}_i \mathrm{m}_i$ as the expectation value of M in the ith eigenstate we can rewrite the last equation as

$$\langle \mathrm{M}(t) \rangle = \sum_i \lambda_i^t \alpha_i m_i.$$

Since we have that in the limit $t \to \infty$ the mean value is dominated by the the largest eigenvalue λ_0, we can rewrite the last equation as

$$\langle \mathrm{M}(t) \rangle = \langle \mathrm{M}(\infty) \rangle + \sum_{i \neq 0} \lambda_i^t \alpha_i m_i.$$

We define the quantity

Time Auto-correlation Function

The quantities τ_i are the correlation times for the system. They control also the auto-correlation function discussed above. The longest correlation time is obviously given by the second largest eigenvalue τ_1, which normally defines the correlation time discussed above. For large times, this is the only correlation time that survives. If higher eigenvalues of the transition matrix are well separated from λ_1 and we simulate long enough, τ_1 may well define the correlation time. In other cases we may not be able to extract a reliable result for τ_1. Coming back to the time correlation function $\phi(t)$ we can present a more general definition in terms of the mean magnetizations $\langle \mathrm{M}(t) \rangle$. Recalling that the mean value is equal to $\langle \mathrm{M}(\infty) \rangle$ we arrive at the expectation values

$$\phi(t) = \langle \mathrm{M}(0) - \mathrm{M}(\infty) \rangle \langle \mathrm{M}(t) - \mathrm{M}(\infty) \rangle,$$

resulting in

$$\phi(t) = \sum_{i,j \neq 0} m_i \alpha_i m_j \alpha_j e^{-t/\tau_i},$$

Correlation Time

If the correlation function decays exponentially

$$\phi(t) \sim \exp\left(-t/\tau\right)$$

then the exponential correlation time can be computed as the average

$$\tau_{\mathrm{exp}} = -\langle \frac{t}{log|\frac{\phi(t)}{\phi(0)}|} \rangle.$$

If the decay is exponential, then

$$\int_0^\infty dt\phi(t) = \int_0^\infty dt\phi(0)\exp\left(-t/\tau\right) = \tau\phi(0),$$

which suggests another measure of correlation

$$\tau_{\mathrm{int}} = \sum_k \frac{\phi(k)}{\phi(0)},$$

called the integrated correlation time

Resampling methods: Blocking

The blocking method was made popular by Flyvbjerg and Pedersen (1989) and has become one of the standard ways to estimate the variance $\mathrm{var}(\widehat{\theta})$ for exactly one estimator $\widehat{\theta}$, namely $\widehat{\theta} = \overline{X}$, the mean value.
Assume $n = 2^d$ for some integer $d > 1$ and $X_1, X_2, \cdots, X_n$ is a stationary time series to begin with. Moreover, assume that the series is asymptotically uncorrelated. We switch to vector notation by arranging $X_1, X_2, \cdots, X_n$ in an n-tuple. Define:

$$\hat{X} = (X_1, X_2, \cdots, X_n).$$

Why blocking?

The strength of the blocking method is when the number of observations, n is large. For large n, the complexity of dependent bootstrapping scales poorly, but the blocking method does not, moreover, it becomes more accurate the larger n is.

Blocking Transformations

We now define the blocking transformations. The idea is to take the mean of subsequent pair of elements from $\boldsymbol{X}$ and form a new vector $\boldsymbol{X}_1$. Continuing in the same way by taking the mean of subsequent pairs of elements of $\boldsymbol{X}_1$ we obtain $\boldsymbol{X}_2$, and so on. Define $\boldsymbol{X}_i$ recursively by:

$$
\begin{aligned}
(\boldsymbol{X}_0)_k &\equiv (\boldsymbol{X})_k \\
(\boldsymbol{X}_{i+1})_k &\equiv \frac{1}{2}\Big((\boldsymbol{X}_i)_{2k-1} + (\boldsymbol{X}_i)_{2k}\Big) \qquad \text{for all} \qquad 1 \leq i \leq d-1
\end{aligned}
\tag{8}
$$

Blocking transformations

The quantity $\boldsymbol{X}_k$ is subject to k **blocking transformations**. We now have d vectors $\boldsymbol{X}_0, \boldsymbol{X}_1, \cdots, \boldsymbol{X}_{d-1}$ containing the subsequent averages of observations. It turns out that if the components of $\boldsymbol{X}$ is a stationary time series, then the components of $\boldsymbol{X}_i$ is a stationary time series for all $0 \leq i \leq d-1$

We can then compute the autocovariance (or just covariance), the variance, sample mean, and number of observations for each i. Let $\gamma_i, \sigma_i^2, \overline{X}_i$ denote the covariance, variance and average of the elements of $\boldsymbol{X}_i$ and let n_i be the number of elements of $\boldsymbol{X}_i$. It follows by induction that $n_i = n/2^i$.

Blocking Transformations

Using the definition of the blocking transformation and the distributive property of the covariance, it is clear that since $h = |i - j|$ we can define

$$\begin{aligned}
\gamma_{k+1}(h) &= cov\left((X_{k+1})_i, (X_{k+1})_j\right) \\
&= \frac{1}{4} cov\left((X_k)_{2i-1} + (X_k)_{2i}, (X_k)_{2j-1} + (X_k)_{2j}\right) \\
&= \frac{1}{2}\gamma_k(2h) + \frac{1}{2}\gamma_k(2h+1)\ \mathrm{h} = 0 \qquad (9) \\
&= \frac{1}{4}\gamma_k(2h-1) + \frac{1}{2}\gamma_k(2h) + \frac{1}{4}\gamma_k(2h+1) \quad \mathrm{else} \qquad (10)
\end{aligned}$$

The quantity $\hat{X}$ is asymptotically uncorrelated by assumption, $\hat{X}_k$ is also asymptotic uncorrelated. Let's turn our attention to the variance of the sample mean $\mathrm{var}(\overline{X})$.

Blocking Transformations, getting there

We have

$$\mathrm{var}(\overline{X}_k) = \frac{\sigma_k^2}{n_k} + \underbrace{\frac{2}{n_k}\sum_{h=1}^{n_k-1}\left(1-\frac{h}{n_k}\right)\gamma_k(h)}_{\equiv e_k} = \frac{\sigma_k^2}{n_k} + e_k \quad \text{if} \quad \gamma_k(0) = \sigma_k^2. \tag{11}$$

The term e_k is called the **truncation error**:

$$e_k = \frac{2}{n_k}\sum_{h=1}^{n_k-1}\left(1-\frac{h}{n_k}\right)\gamma_k(h). \tag{12}$$

We can show that $\mathrm{var}(\overline{X}_i) = \mathrm{var}(\overline{X}_j)$ for all $0 \leq i \leq d-1$ and $0 \leq j \leq d-1$.

Blocking Transformations, final expressions

We can then wrap up

$$
\begin{aligned}
n_{j+1}\overline{X}_{j+1} &= \sum_{i=1}^{n_{j+1}} (\hat{X}_{j+1})_i = \frac{1}{2}\sum_{i=1}^{n_j/2} (\hat{X}_j)_{2i-1} + (\hat{X}_j)_{2i} \\
&= \frac{1}{2}\left[(\hat{X}_j)_1 + (\hat{X}_j)_2 + \cdots + (\hat{X}_j)_{n_j}\right] = \underbrace{\frac{n_j}{2}}_{=n_{j+1}} \overline{X}_j = n_{j+1}\overline{X}_j.
\end{aligned}
\tag{13}
$$

By repeated use of this equation we get $\mathrm{var}(\overline{X}_i) = \mathrm{var}(\overline{X}_0) = \mathrm{var}(\overline{X})$ for all $0 \leq i \leq d-1$. This has the consequence that

$$
\mathrm{var}(\overline{X}) = \frac{\sigma_k^2}{n_k} + e_k \qquad \text{for all} \qquad 0 \leq k \leq d-1. \tag{14}
$$

More on the blocking method

Flyvbjerg and Petersen demonstrated that the sequence $\{e_k\}_{k=0}^{d-1}$ is decreasing, and conjecture that the term e_k can be made as small as we would like by making k (and hence d) sufficiently large. The sequence is decreasing. It means we can apply blocking transformations until e_k is sufficiently small, and then estimate $\mathrm{var}(\overline{X})$ by $\widehat{\sigma}_k^2/n_k$.

For an elegant solution and proof of the blocking method, see the recent article of Marius Jonsson (former MSc student of the Computational Physics group).

Example code form last week

```
# 2-electron VMC code for 2dim quantum dot with importance sampling
# Using gaussian rng for new positions and Metropolis- Hastings
# Added energy minimization
from math import exp, sqrt
from random import random, seed, normalvariate
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
from matplotlib import cm
from matplotlib.ticker import LinearLocator, FormatStrFormatter
from scipy.optimize import minimize
import sys
import os

# Where to save data files
PROJECT_ROOT_DIR = "Results"
DATA_ID = "Results/EnergyMin"

if not os.path.exists(PROJECT_ROOT_DIR):
    os.mkdir(PROJECT_ROOT_DIR)

if not os.path.exists(DATA_ID):
    os.makedirs(DATA_ID)

def data_path(dat_id):
    return os.path.join(DATA_ID, dat_id)

outfile = open(data_path("Energies.dat"),'w')
```

Resampling analysis

The next step is then to use the above data sets and perform a resampling analysis using the blocking method The blocking code, based on the article of Marius Jonsson is given here

```
# Common imports
import os

# Where to save the figures and data files
DATA_ID = "Results/EnergyMin"

def data_path(dat_id):
    return os.path.join(DATA_ID, dat_id)

infile = open(data_path("Energies.dat"),'r')

from numpy import log2, zeros, mean, var, sum, loadtxt, arange, array,
from numpy.linalg import inv

def block(x):
    # preliminaries
    n = len(x)
    d = int(log2(n))
    s, gamma = zeros(d), zeros(d)
    mu = mean(x)

    # estimate the auto-covariance and variances
    # for each blocking transformation
    for i in arange(0,d):
```